Dhruv Mishra

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Summary

Software Engineer at Microsoft building production **AI agents** and the **Azure** backend services they run on. Shipped MCP-based incident-triage agents that cut manual triage by 40% on a **7B+ requests/day** platform, reduced its Azure spend by $240k/year, and sandboxed native code to ship enterprise encryption in Microsoft Office.

Experience

Software Engineer Jun. 2024 – Present
Microsoft *Noida, India*

- Built and shipped production **LLM incident-triage agents** for on-call engineers: **MCP** servers give the agent Kusto, service-health, and config tools through structured tool calls, with fallback logic and human escalation. Cut manual low/mid-severity triage workload by **40%**, measured by replaying previously resolved incidents with known mitigations and confirmed by developer usage.
- Cut Azure spend by **$240k/year (15%)** on a 7B+ requests/day Service Fabric backend by raising **Redis** cache hit ratio from 31% to 80%, tuning Cosmos DB and log retention, and retuning autoscaling across 12+ VM Scale Sets, within the 99.99% availability target; impact measured over 3 months of Geneva/Kusto telemetry.
- Shipped PDF encryption for Word, Excel, and PowerPoint in Office for the web by sandboxing a C++ encryption SDK; **reduced attack surface** by removing direct network access from native code, constraining its execution, and routing Azure Rights Management Service (RMS) calls through a C# proxy over IPC.
- Cut Office Android cold-start and file-open latency by ~**500ms** with custom baseline profiles for Word, Excel, and PowerPoint, and Excel's loading-shimmer time by **99% (300ms → 3ms)**; validated through staged-rollout telemetry.
- Primary maintainer of **Fluent UI Android**, Microsoft's open-source Jetpack Compose library; shipped 10 releases used by Copilot, Office, Outlook, OneDrive, and Teams, and owned its security compliance.

Machine Learning Engineer Feb. 2024 – Jun. 2024
growIndigo *Delhi, India*

- Replaced a manual Google Earth Engine crop-classification workflow with a Python XGBoost + Random Forest pipeline, improving accuracy from 80% to 93%.

Software Engineering Intern May 2023 – Jul. 2023
Microsoft *Noida, India*

- Built a **TypeScript** connection-management backend for Microsoft Loop that syncs Jira, Trello, and GitHub connection state; later reused as the shared backend for Loop third-party integrations.
- Integrated GitHub and Azure DevOps **REST APIs** into the Microsoft Power Platform Connector and raised module test coverage to **over 60%**.

Projects

AI Portfolio with Live Voice Agent | whoisdhruv.com | *Next.js, React, TypeScript, Gemini Live, RAG, Docker*
Production personal site with a **voice agent** visitors can talk to and interrupt mid-sentence (full-duplex audio over WebSocket). The agent uses 25 tools to navigate pages, fill forms, and run commands for the user. Also has a **RAG chatbot** that answers questions about my experience and projects, with 5 selectable LLMs and a self-hosted MoE LLM as fallback. Deployed with Docker and GitHub Actions.

Invoke – Voice-Controlled Coding Agents | github.com/Dhruv-Mishra/Invoke-Voice | *Electron, Node.js, Python, llama.cpp, MCP*
Windows voice assistant that answers simple requests itself and **hands complex coding tasks to GitHub Copilot CLI agents**, each in its own Git worktree. Speech-to-text, the LLM, and text-to-speech all **run locally** (Parakeet, quantized Gemma/Qwen on llama.cpp, Kokoro). **Speculative decoding** with a draft model plus thread tuning made generation **35%** faster; a benchmark harness picked a speech model with **57%** fewer errors than Whisper Small at 5x the speed.

Jarvis – Voice AI Dispatch Agent | jarvis.whoisdhruv.com | *Next.js, Python, MCP, RAG*
Voice agent for a logistics app that makes calls, negotiates freight rates, and submits quotes, using an LLM planner with MCP tools and RAG over load and carrier data.

Technical Skills

Languages: C#, TypeScript, Python, Kotlin, C++, Java, JavaScript, SQL
AI & ML: Agentic AI, LLM agents, multi-agent systems, MCP, tool calling, RAG, embeddings, LLM evaluation, prompt engineering, speculative decoding, quantization (GGUF), llama.cpp, vLLM, voice agents (STT/TTS), multimodal LLMs, classical ML (XGBoost), OpenCV
Backend & Cloud: Azure (Service Fabric, VM Scale Sets, Cosmos DB, RMS), Redis, microservices, Node.js, REST APIs, Docker, Nginx, Linux, Cloudflare, GCP, OCI
Frontend & Mobile: React, Next.js, Vue, Electron, Tailwind CSS, Android, Jetpack Compose
Tools & Practices: Git, GitHub Actions, Azure DevOps, Kusto (ADX), CI/CD, observability, distributed systems, system design

Education

Indraprastha Institute of Information Technology (IIIT-Delhi) Delhi, India
B.Tech (Honors), Computer Science and Applied Mathematics – CGPA 8.96 / 10.0 *Dec. 2020 – May 2024*

- Relaxed Concurrent Counting Bloom Filters: researcher at the Distributed Computing Lab under Prof. Bapi Chatterjee; built a lock-free C++ Counting Bloom Filter with **3x** throughput over the baseline under high contention.
- Course Recommendation System: Python NLP similarity-scoring recommender built under Prof. Dhruv Kumar.

Achievements

Competitive Programming: Codeforces Expert (1703) | CodeChef 5-star (2003) | Code Jam 2023 Round A #291 | Kick Start 2022 Round H #353
Awards: Microsoft Hackathon 2026 Most Voted ("Shell Assistant", incident-response agent); Microsoft Hackathon 2025 Fundamental Category ("WXP ActionEase")
Leadership: AI Champ, Microsoft AI Productivity team; ran workshops on multi-agent engineering and LLM automation